

Publications

Katrín Björnsdóttir [Google Scholar profile](#)

Peer-reviewed articles

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- 2026 Verdonen M, Barrio IC, Barbero-Palacios L, ... **Björnsdóttir K**, ... Myers-Smith IH (28 authors). **Borealization of tundra ecosystems with climate and land-use change.** *Environmental Research Letters* 21(1): 013002. [doi:10.1088/1748-9326/ae2e18](https://doi.org/10.1088/1748-9326/ae2e18)
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- 2025 Barrio IC, Vuorinen KEM, Barbero-Palacios L, ... **Björnsdóttir K**, ... Kamenova S (79 authors). **Emerging priorities in terrestrial herbivory research in the Arctic.** *Arctic Science* 11. [doi:10.1139/as-2024-0080](https://doi.org/10.1139/as-2024-0080)
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- Schwieger S, Dorrepaal E, Petit Bon M, ... **Björnsdóttir K**, ... Sarneel JM (42 authors). **Environmental conditions modulate warming effects on plant litter decomposition globally.** *Ecology Letters* 28(1): e70026. [doi:10.1111/ele.70026](https://doi.org/10.1111/ele.70026)
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- Eidesen PB, Brysting AK, Hagen KR, ... **Björnsdóttir K**, ... Volden IK (26 authors). **Ecological and evolutionary consequences of ploidy-driven trait variation: insights from *Saxifraga oppositifolia* L.** *Arctic Science* 11. [doi:10.1139/as-2024-0020](https://doi.org/10.1139/as-2024-0020)
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- 2024 Sarneel JM, Hefting MM, Sandén T, ... **Björnsdóttir K**, ... Keuskamp JA (100 authors). **Reading tea leaves worldwide: decoupled drivers of initial litter decomposition mass-loss rate and stabilization.** *Ecology Letters* 27(5): e14415. [doi:10.1111/ele.14415](https://doi.org/10.1111/ele.14415)
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- Barbero-Palacios L, Barrio IC, García Criado M, ... **Björnsdóttir K**, ... Soininen EM (36 authors). **Herbivore diversity effects on Arctic tundra ecosystems: a systematic review.** *Environmental Evidence* 13: 6. [doi:10.1186/s13750-024-00330-9](https://doi.org/10.1186/s13750-024-00330-9)
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- 2023 Vandvik V, Halbritter AH, Althuizen IHJ, ... **Björnsdóttir K**, ... Enquist BJ (35 authors). **Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard.** *Scientific Data* 10: 578. [doi:10.1038/s41597-023-02467-7](https://doi.org/10.1038/s41597-023-02467-7)
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- Jónsdóttir IS, Halbritter AH, Christiansen CT, Althuizen IHJ, Haugum SV, Henn JJ, **Björnsdóttir K**, Maitner BS, Malhi Y, Michaletz ST, Roos RE, Klanderud K, Lee H, Enquist BJ, Vandvik V. **Intraspecific trait variability is a key feature underlying high Arctic plant community resistance to climate warming.** *Ecological Monographs* 93(1): e1555. [doi:10.1002/ecm.1555](https://doi.org/10.1002/ecm.1555)
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- 2022 Lembrechts JJ, van den Hoogen J, Aalto J, ... **Björnsdóttir K**, ... Lenoir J. **Global maps of soil temperature.** *Global Change Biology* 28(9): 3110–3144. [doi:10.1111/gcb.16060](https://doi.org/10.1111/gcb.16060)
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- 2021 Soininen EM, Barrio IC, Björkås R, Björnsdóttir K, ... Speed JDM (44 authors). **Location of studies and evidence of effects of herbivory on Arctic vegetation: a systematic map.** *Environmental Evidence* 10: 25. [doi:10.1186/s13750-021-00240-0](https://doi.org/10.1186/s13750-021-00240-0)
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- Björnsdóttir K, Barrio IC, Jónsdóttir IS. **Long-term warming manipulations reveal complex decomposition responses across different tundra vegetation types.** *Arctic Science* 8(3): 979–991. [doi:10.1139/as-2020-0046](https://doi.org/10.1139/as-2020-0046)
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- 2019 Mulloy TA, Barrio IC, Björnsdóttir K, Jónsdóttir IS, Hik DS. **Fertilisers mediate the short-term effects of sheep grazing in the Icelandic highlands.** *Icelandic Agricultural Sciences* 32: 75–85. [doi:10.16886/ias.2019.07](https://doi.org/10.16886/ias.2019.07)
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Manuscripts

- 2026 Björnsdóttir K, Myers-Smith IH, Thomas H, Pacheco-Riaño LC, Elmendorf SC, Wårlind D, Miller PA, Petit Bon M, Björk RG, Cornelissen JHC, Lang SI, Jónsdóttir IS, Vandvik V, ... Bjorkman A. **Plant community litter decomposability increases with warming across the tundra.** *Manuscript ready for submission.*
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- Björnsdóttir K, Gya R, Bjorkman A. **Temperature sensitivity of tundra plant functional traits to experimental warming.** *Manuscript in preparation.*
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- 2025 Borderieux J, Thomas H, Myers-Smith IH, Høye T, Petit Bon M, Lembrechts J, Walker E, Björnsdóttir K, ... Sarneel J (40 authors). **Litter quality outweighs climate as a driver of decomposition across the tundra biome.** *In revision, Functional Ecology. Preprint: EcoEvoRxiv.* [doi:10.32942/x2q352](https://doi.org/10.32942/x2q352)
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Theses

- 2026 **PhD thesis, University of Gothenburg**
- *Consequences of climate change for ecosystem functioning mediated by plant responses across the tundra.* Department of Biological and Environmental Sciences. hdl.handle.net/2077/93153
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- 2018 **M.Sc. thesis, University of Iceland**
- *Decomposition responses to climate warming and sheep grazing in the high and sub-Arctic.* Advisors: Ingibjörg Svala Jónsdóttir and Isabel C. Barrio. hdl.handle.net/1946/30518
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- 2015 **B.Sc. thesis, University of Iceland**
- *Áhrif vegagerðar í Skaftáreldahrauni á landnám gulvíðis, loðvíðis og krækilyngs.*